
MATH 241 Engineering Mathematics I

Problem Set-IV Solutions

Problem 1

Define m -dimensional vector space $\mathbf{F} = \mathbf{V} - \mathbf{W}$, where $m = n - k$. Since $\mathbf{W}$ is a k -dimensional subspace of $\mathbf{V}$, the vector space $\mathbf{F}$ is also an m -dimensional subspace of $\mathbf{V}$.

Let us assume that there is no such vector $\mathbf{w} \in \mathbf{W}$ which can be written as a linear combination of $\{v_{i1}, v_{i2}, \dots, v_{im}\}$. The fact that all $\{v_{i1}, v_{i2}, \dots, v_{im}\}$ vectors lie in the $\mathbf{F}$ space is a contradiction to the given condition ($m > n - k$). Therefore, our assumption is wrong because there exists at least one such vector $\mathbf{w} \in \mathbf{W}$ which can be written as a linear combination of $\{v_{i1}, v_{i2}, \dots, v_{im}\}$.

Problem 2

Let

$$M_2(\mathbb{R}) = \left\{ \begin{bmatrix} x & y \\ z & t \end{bmatrix} \mid x, y, z, t \in \mathbb{R} \right\}$$

We can write elements of $M_2(\mathbb{R})$ as follows

$$\begin{cases} x = a + c \\ y = b + d \\ z = d - b \\ t = a - c \end{cases} \Rightarrow \begin{cases} a = \frac{x+t}{2} \\ b = \frac{y-z}{2} \\ c = \frac{x-t}{2} \\ d = \frac{y+z}{2} \end{cases}$$

Therefore,

$$M_2(\mathbb{R}) = \begin{bmatrix} a & b \\ -b & a \end{bmatrix} + \begin{bmatrix} c & d \\ d & -d \end{bmatrix}$$

Suppose that the elements of $M_2(\mathbb{R})$ are not unique, i.e., it has two different representatives:

$$\begin{aligned} M_2(\mathbb{R}) &= \begin{bmatrix} a_1 & b_1 \\ -b_1 & a_1 \end{bmatrix} + \begin{bmatrix} c_1 & d_1 \\ d_1 & -c_1 \end{bmatrix} \\ M_2(\mathbb{R}) &= \begin{bmatrix} a_2 & b_2 \\ -b_2 & a_2 \end{bmatrix} + \begin{bmatrix} c_2 & d_2 \\ d_2 & -c_2 \end{bmatrix} \end{aligned}$$

We have

$$\begin{cases} x = a_1 + c_1 = a_2 + c_2 \\ y = b_1 + d_1 = b_2 + d_2 \\ z = d_1 - b_1 = d_2 - b_2 \\ t = a_1 - c_1 = a_2 - c_2 \end{cases}$$

Solving the above equations, we can see that

$$\begin{cases} a_1 = a_2 \\ b_1 = b_2 \\ c_1 = c_2 \\ d_1 = d_2 \end{cases}$$

which means that the representations are unique. Thus,

$$M_2(\mathbb{R}) = W_1 \oplus W_2.$$

Let $x \in M_n(\mathbb{R})$. We know that any $n \times n$ matrix can be written based on the summation of a symmetric and a skew-symmetric matrix as

$$x = \frac{1}{2}(x + x^T) + \frac{1}{2}(x - x^T)$$

Now, let $x = s_1 + k_1$ and $x = s_2 + k_2$, where s_1 and s_2 are symmetric and k_1 and k_2 are skew-symmetric. Thus, $x^T = s_1 - k_1 = s_2 - k_2$ and $x^T = s_1 + k_1 = s_2 + k_2$. It can be shown that $s_1 = s_2$ and $k_1 = k_2$. Consequently, the representation of x as the sum of a skew-symmetric and a symmetric matrices is unique.

Problem 3

$$\begin{aligned}
 V &= \left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \in \mathbb{R}^4 \mid x_1 = x_3 + x_4; x_2 = x_3 - x_4 \right\} \\
 &= \left\{ \mathbf{x} = \begin{bmatrix} x_3 + x_4 \\ x_3 - x_4 \\ x_3 \\ x_4 \end{bmatrix} \in \mathbb{R}^4 \right\} \\
 &= \left\{ \mathbf{x} = x_3 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\} \\
 &= \text{SPAN} \{ \mathbf{e}_1, \mathbf{e}_2 \}
 \end{aligned}$$

where

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix} \quad (1)$$

Since $\mathbf{e}_1 \cdot \mathbf{e}_2 = 0$, $\mathbf{e}_1$ and $\mathbf{e}_2$ form orthogonal bases.

By normalizing the above vectors, the orthonormal bases of $\mathbf{V}$ are obtained as

$$\bar{\mathbf{e}}_1 = \frac{\mathbf{e}_1}{\|\mathbf{e}_1\|} = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ 0 \end{bmatrix}, \quad \bar{\mathbf{e}}_2 = \frac{\mathbf{e}_2}{\|\mathbf{e}_2\|} = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \\ 0 \\ \frac{1}{\sqrt{3}} \end{bmatrix}$$

Now, to find orthonormal bases for $\mathbf{V}^\perp$ ($\bar{\mathbf{e}}_3$ and $\bar{\mathbf{e}}_4$), just find corresponding orthogonal vectors to $\mathbf{e}_1$ and $\mathbf{e}_2$.

$$\begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} e_{y_1} \\ e_{y_2} \\ e_{y_3} \\ e_{y_4} \end{bmatrix} = 0, \quad \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} e_{y_1} \\ e_{y_2} \\ e_{y_3} \\ e_{y_4} \end{bmatrix} = 0$$

By solving the above equations, we get

$$\begin{aligned}
 e_{y_1} &= 1/2e_{y_4} - 1/2e_{y_3} \\
 e_{y_2} &= -1/2e_{y_4} - 1/2e_{y_3}
 \end{aligned}$$

So,

$$\mathbf{e}_3 = \begin{bmatrix} -1/2 \\ 1/2 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{e}_4 = \begin{bmatrix} -1/2 \\ -1/2 \\ 1 \\ 0 \end{bmatrix}$$

Again, to have orthonormal bases, we normalize these vectors as

$$\bar{\mathbf{e}}_3 = \frac{\mathbf{e}_3}{\|\mathbf{e}_3\|} = \begin{bmatrix} -\sqrt{1/6} \\ \sqrt{1/6} \\ 0 \\ \sqrt{2/3} \end{bmatrix}, \quad \bar{\mathbf{e}}_4 = \frac{\mathbf{e}_4}{\|\mathbf{e}_4\|} = \begin{bmatrix} -\sqrt{1/6} \\ -\sqrt{1/6} \\ \sqrt{2/3} \\ 0 \end{bmatrix}$$

Problem 4

Given that S is the subset of inner product space V .

- a) $S \cap S^\perp = \{\mathbf{0}\} \Rightarrow \text{True}$
- b) $S \subset (S^\perp)^\perp \Rightarrow \text{True}$
- c) $S = (S^\perp)^\perp \Rightarrow \text{False}$
- d) $(S^\perp)^\perp = \text{SPAN}(S) \Rightarrow \text{True}$
- e) $[(S^\perp)^\perp]^\perp = S^\perp \Rightarrow \text{True}$
- f) $S^\perp$ is always a subspace of $V \Rightarrow \text{True}$
- g) $S^\perp \oplus \text{SPAN}(S) = V \Rightarrow \text{True}$
- h) $\dim(S^\perp) + \dim((S^\perp)^\perp) = \dim(V) \Rightarrow \text{True}$

a) If S is a subset of an inner product space V , then $S^\perp$ is the set of all vectors orthogonal to the vectors in S . Now, let u be a vector in $S \cap S^\perp$, i.e., u is orthogonal to all vectors in S including itself. Thus, $u = \mathbf{0} \Rightarrow S \cap S^\perp = \{\mathbf{0}\}$.

b) $S^\perp$ is the set of all vectors orthogonal to the vectors in the subset S . Similarly, $(S^\perp)^\perp$ is the set of all vectors orthogonal to vectors in $S^\perp$. Since S is not a subspace, $(S^\perp)^\perp = S + \{\text{some other vectors orthogonal to } S^\perp\}$. Therefore, $S \subset (S^\perp)^\perp$.

c) As evident from part b, $S = (S^\perp)^\perp$ is valid only if S is a subspace of V .

d) Since $S \subset (S^\perp)^\perp$, every vector in $(S^\perp)^\perp$ can be written as linear combination of vectors in S even though they are not LI. Therefore, $(S^\perp)^\perp = \text{SPAN}(S)$

e) $S^\perp$ contains all vectors orthogonal to vectors in S . Let $u \in S$, $x_1 \in S^\perp$, and $x_2 \in S^\perp$, then

$$(ax_1 + bx_2) \cdot u = ax_1 \cdot u + bx_2 \cdot u = 0.$$

Therefore, $ax_1 + bx_2$ also belongs to $S^\perp$.

Let $W = S^\perp$ be a subspace of V . Then, $w \in W$ will be orthogonal to any vector $u \in W^\perp$. Now, let $v \in (W^\perp)^\perp$ is a vector orthogonal to every vector in $W^\perp$. If we can show that there does not exist any nonzero u satisfies $v = u + w$, it is proved that $[(S^\perp)^\perp]^\perp = S^\perp$. We have

$$v \cdot u = 0 \Rightarrow (u + w) \cdot u = 0 \Rightarrow u \cdot u + w \cdot u = 0 \Rightarrow u \cdot u = 0 \Rightarrow u = \mathbf{0} \Rightarrow v = w$$

It means that for every vector in W , there is one identical vector in $(W^\perp)^\perp$. Consequently, we have

$$\begin{aligned} (W^\perp)^\perp &= W \\ ((S^\perp)^\perp)^\perp &= S^\perp \end{aligned}$$

f) See part e.

g) For any inner product space V ,

$$V = U \oplus W$$

where U and W are orthogonal complements of each other. Thus,

$$\begin{aligned} V &= S^\perp \oplus (S^\perp)^\perp \\ \text{from part e } \Rightarrow V &= S^\perp \oplus \text{SPAN}(S) \end{aligned}$$

h) From part g, we have

$$V = S^\perp \oplus (S^\perp)^\perp$$

Then,

$$\dim(V) = \dim(S^\perp) \oplus \dim((S^\perp)^\perp)$$

Problem 5

Given that

$$x_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \end{bmatrix}, x_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, x_3 = \begin{bmatrix} 4 \\ 3 \\ 4 \\ -1 \end{bmatrix}, x_4 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix},$$

we can see that $x_3 = x_1 + 3x_2$. Thus, x_3 is linearly dependant on x_1 and x_2 .

Using Gram Schmidt procedure, v_3 should be zero, i.e., orthogonal basis v_1, v_2, v_4 are sufficient to span V .

Let's apply the procedure

$$v_1 = x_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

$$v_2 = x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} v_1 = \begin{bmatrix} 1/3 \\ 1 \\ 1/2 \\ 2/3 \end{bmatrix}$$

$$v_3 = x_3 - \frac{x_3 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_3 \cdot v_2}{v_2 \cdot v_2} v_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$v_4 = x_4 - \frac{x_4 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_4 \cdot v_2}{v_2 \cdot v_2} v_2 - \frac{x_4 \cdot v_3}{v_3 \cdot v_3} v_3 = \begin{bmatrix} 3/5 \\ -1/5 \\ -2/5 \\ 1/5 \end{bmatrix}$$

Therefore,

$$\{v_1, v_2, v_4\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1/3 \\ 1 \\ 1/2 \\ 2/3 \end{bmatrix}, \begin{bmatrix} 3/5 \\ -1/5 \\ -2/5 \\ 1/5 \end{bmatrix} \right\}.$$

form an orthonormal basis.

Problem 6

$A_{m \times n} = [a_1, \dots, a_n]$ forming n LI vectors. The GramSchmidt procedure states that

$$\begin{aligned}
 v_1 &= a_1 \\
 v_2 &= a_2 - \frac{a_2 \cdot v_1}{v_1 \cdot v_1} v_1 \\
 &\vdots \\
 v_i &= a_i - \frac{a_i \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{a_i \cdot v_2}{v_2 \cdot v_2} v_2 - \dots - \frac{a_i \cdot v_{i-1}}{v_{i-1} \cdot v_{i-1}} v_{i-1} \\
 \Rightarrow a_i &= v_i + \frac{a_i \cdot v_1}{v_1 \cdot v_1} v_1 + \frac{a_i \cdot v_2}{v_2 \cdot v_2} v_2 + \dots - \frac{a_i \cdot v_{i-1}}{v_{i-1} \cdot v_{i-1}} v_{i-1}
 \end{aligned}$$

Let $q_i = \frac{v_i}{\|v_i\|} \Rightarrow a_i = \|v_i\| q_i + (a_i \cdot q_1) q_1 + \dots + (a_i \cdot q_{i-1}) q_{i-1}$

$$\begin{aligned}
 \Rightarrow a_1 &= \|v_1\| q_1 \\
 \Rightarrow a_2 &= \|v_2\| q_2 + (a_2 \cdot q_1) q_1 \\
 &\vdots \\
 \Rightarrow a_n &= \|v_n\| q_n + (a_n \cdot q_1) q_1 + \dots + (a_n \cdot q_{n-1}) q_{n-1}
 \end{aligned}$$

Rewriting the $a_1, a_2, \dots, a_n$ in matrix form, we have

$$\begin{aligned}
 [a_1, a_2, \dots, a_n] &= [q_1, q_2, \dots, q_n] \begin{bmatrix} \|v_1\| & a_2 \cdot q_1 & a_3 \cdot q_1 & a_4 \cdot q_1 & \dots & a_n \cdot q_1 \\ 0 & \|v_2\| & a_3 \cdot q_2 & a_4 \cdot q_2 & \dots & a_n \cdot q_2 \\ 0 & 0 & \|v_3\| & a_4 \cdot q_3 & \dots & a_n \cdot q_3 \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & 0 & \dots & \|v_n\| \end{bmatrix} \\
 &= [q_1, q_2, \dots, q_n] \begin{bmatrix} r_{11} & r_{12} & r_{13} & r_{14} & \dots & r_{1n} \\ 0 & r_{22} & r_{23} & r_{24} & \dots & r_{2n} \\ 0 & 0 & r_{33} & r_{43} & \dots & r_{n3} \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & 0 & \dots & r_{nn} \end{bmatrix} \\
 &= Q \cdot R.
 \end{aligned}$$

This is called QR decomposition.

Problem 7

Using Q-point formula we have

$$x - 15 = \frac{25 - 15}{3 - 1}(t - 1) \Rightarrow x = 10 + 5t$$

with $x_0 = 10$ and $v_0 = 5$

Now, taking $(t_1, x_1) = (1, 16)$ and $(t_2, x_2) = (3, 22)$,

$$x - 16 = \frac{22 - 16}{3 - 1}(t - 1) \Rightarrow x = 13 + 3t$$

with $x_0 = 13$ and $v_0 = 3$

Taking $(1, 16)$, $(3, 22)$, $(5, 39)$ and using $x = x_0 + t.v_0$, we have

$$\begin{aligned} \begin{cases} 16 = x_0 + v_0 \\ 22 = x_0 + 3v_0 \\ 39 = x_0 + 5v_0 \end{cases} &\Rightarrow \begin{bmatrix} 1 & 1 \\ 1 & 3 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} x_0 \\ v_0 \end{bmatrix} = \begin{bmatrix} 16 \\ 22 \\ 39 \end{bmatrix} \\ &\Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 1 & 3 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 3 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} x_0 \\ v_0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 3 & 5 \end{bmatrix} \begin{bmatrix} 16 \\ 22 \\ 39 \end{bmatrix} \\ &\Rightarrow \begin{bmatrix} 3 & 9 \\ 9 & 35 \end{bmatrix} \begin{bmatrix} x_0 \\ v_0 \end{bmatrix} = \begin{bmatrix} 77 \\ 277 \end{bmatrix} \\ &\Rightarrow \begin{bmatrix} x_0 \\ v_0 \end{bmatrix} = \begin{bmatrix} 3 & 9 \\ 9 & 35 \end{bmatrix}^{-1} \begin{bmatrix} 77 \\ 277 \end{bmatrix} = \begin{bmatrix} 8.42 \\ 5.75 \end{bmatrix} \end{aligned}$$

Thus, the best approximation for given data points is the line

$$x = 8.42 + 5.75t$$